

Æfingaverkefni 1

Taflan:

2^{12}	2^{11}	2^{10}	2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
4096	2048	1024	512	256	128	64	32	16	8	4	2	1

Breytið eftirfarandi tvíundartölum í tugakerfistölur:

$101_2 = 5_{10}$
 $110_2 = 6_{10}$
 $111_2 = 7_{10}$
 $1001_2 = 9_{10}$
 $1100_2 = 12_{10}$
 $1111_2 = 15_{10}$
 $11100_2 = 28_{10}$
 $10010_2 = 18_{10}$
 $100100_2 = 36_{10}$
 $110011_2 = 51_{10}$
 $100010_2 = 34_{10}$
 $111111_2 = 63_{10}$
 $1100110_2 = 102_{10}$
 $1110001_2 = 113_{10}$
 $10000011_2 = 131_{10}$
 $11010000_2 = 208_{10}$

Breytið eftirfarandi tugakerfistölum í tvíundarkerfið:

$14_{10} = 1110_2$
 $21_{10} = 10101_2$
 $27_{10} = 11011_2$
 $31_{10} = 11111_2$
 $42_{10} = 101010_2$
 $47_{10} = 101111_2$
 $52_{10} = 110100_2$
 $59_{10} = 111011_2$
 $63_{10} = 111111_2$
 $66_{10} = 1000010_2$
 $127_{10} = 1111111_2$
 $128_{10} = 10000000_2$
 $168_{10} = 10101000_2$
 $172_{10} = 10101100_2$
 $192_{10} = 11000000_2$
 $255_{10} = 11111111_2$
 $340_{10} = 101010100_2$
 $513_{10} = 1000000001_2$
 $1048_{10} = 10000011000_2$